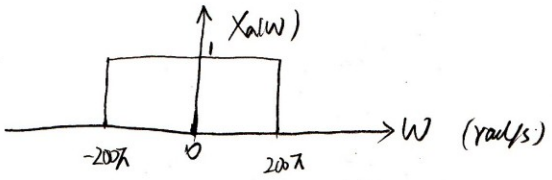
HW #8

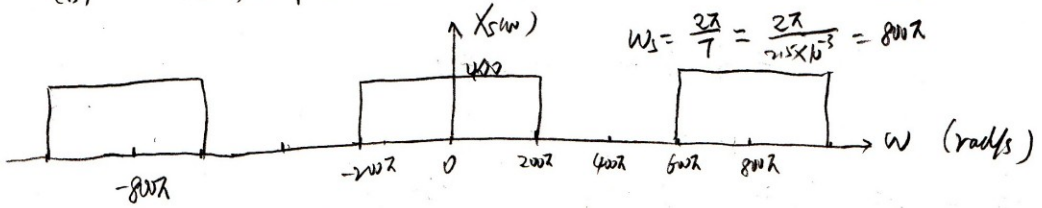
1

1.

(a)

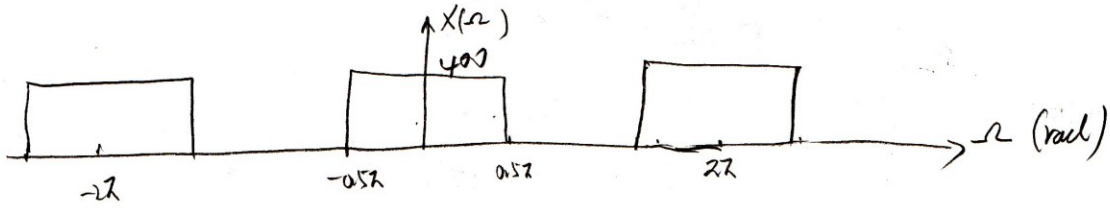


(b) $X_s(w) = \frac{1}{T}\sum X_a(w - n\cdot\frac{2\pi}{T})$ $\quad T = 2.5ms \quad \frac{1}{T} = \frac{1}{2.5\times10^{-3}} = 400$

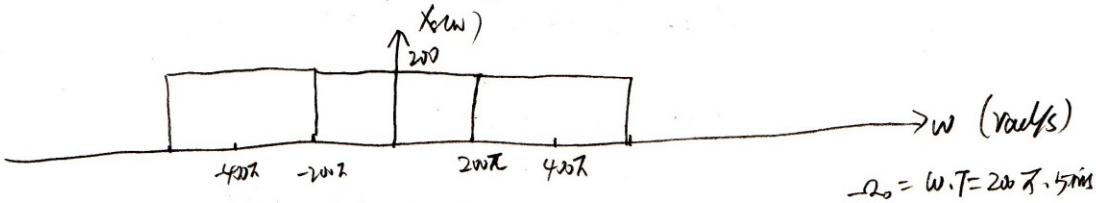


$X(\Omega) = X_s(\frac{\Omega}{T})$ $\quad \Omega_s = w_s\cdot T = 800\pi\cdot 2.5ms = 2\pi$

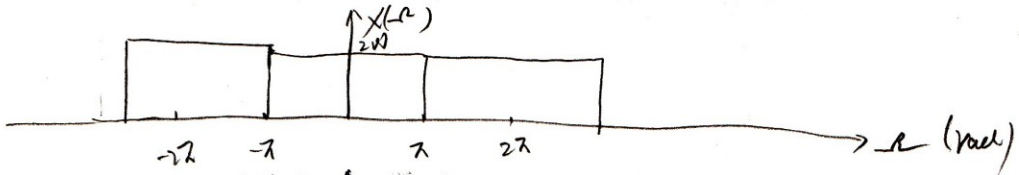
$\Omega_0 = w_0\cdot T = 200\pi\cdot 2.5ms = 0.5\pi$



(c) $T = 5ms$ $\quad w_s = \frac{2\pi}{T} = \frac{2\pi}{5\times10^{-3}} = 400\pi$ $\quad \frac{1}{T} = 200$

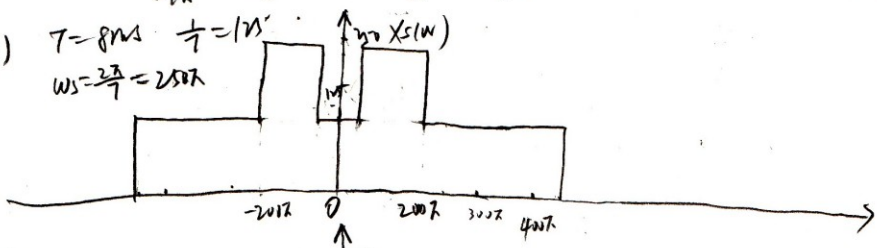


$\Omega_0 = w_0\cdot T = 200\pi\cdot 5ms = \pi$

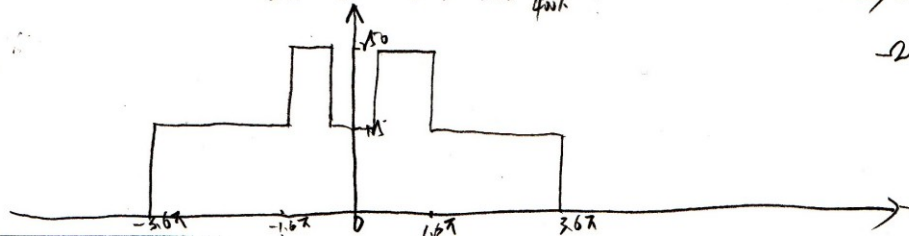


(d) $T = 8ms$ $\quad \frac{1}{T} = 125$

$w_s = \frac{2\pi}{T} = 250\pi$



$\Omega_0 = w_0\cdot T = 200\pi\cdot 8ms = 1.6\pi$



(d) In order to avoid aliasing, $\omega_s \geq 2\omega_0$

$\omega_s \geq 400\pi$

$T_s \leq \frac{2\pi}{\omega_s} \leq \frac{2\pi}{400\pi} = 5ms$

so that $T_{max} = 5ms$

(2)

2. $x(n) = x_a(nT) \Longrightarrow X(\Omega) = \frac{1}{T}\sum_n X_a\left(\frac{\Omega}{T} - n\cdot\frac{2\pi}{T}\right)$

Proof:

let $p(t) = \sum_{n=-\infty}^{\infty}\delta(t-nT) = \frac{1}{T}\sum_{n=-\infty}^{\infty} 1\cdot e^{jn\omega_s t}$ $\omega_s = \frac{2\pi}{T}$

the Fourier Transform (FT) of $p(t)$

$$P(\omega) = \frac{2\pi}{T}\sum_{n=-\infty}^{\infty}\delta(\omega - n\omega_s)$$

So that: $x(t)\cdot p(t) \Longleftrightarrow \frac{1}{2\pi}[X_a(\omega)\otimes P(\omega)]$

$$x_s(t) = x(t)\cdot p(t) \Longleftrightarrow X_s(\omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a(\omega - n\omega_s)$$

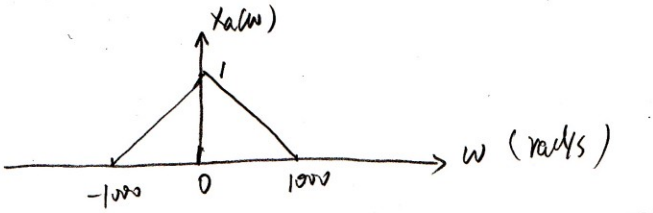
Next: the FT of $x_s(t)$

$$X_s(\omega) = \int_{-\infty}^{\infty} x_s(t)e^{-j\omega t}dt = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega nT}$$

and $$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\Omega n}$$

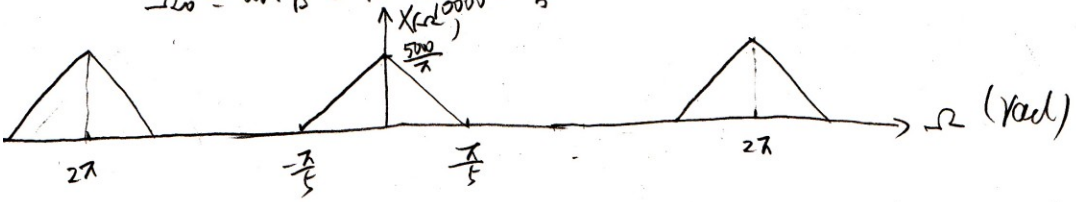
so that $$X(\Omega) = X_s\left(\frac{\Omega}{T}\right) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a\left(\frac{\Omega}{T} - n\cdot\frac{2\pi}{T}\right).$$

3. (a)



$w_s = 10{,}000 \qquad T_s = \frac{2\pi}{w_s} = \frac{2\pi}{10000} \qquad \frac{1}{T_s} = \frac{10000}{2\pi} = \frac{5000}{\pi}$

$\Omega_0 = w_0 \cdot T_s = 1000 \cdot \frac{2\pi}{10000} = \frac{\pi}{5}$



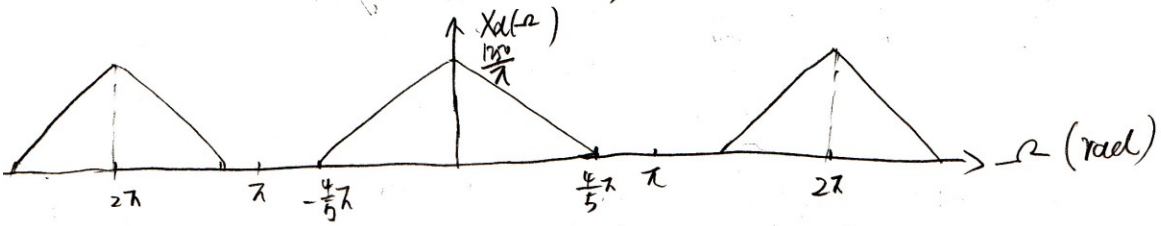
(b). $X_d(\Omega) = \frac{1}{M} X\left(\frac{\Omega}{M}\right)$

The largest value that can not introduce aliasing, with $\Omega_0^d = \pi$

$$M = \frac{\Omega_0^d}{\Omega_0} = \frac{\pi}{\frac{\pi}{5}} = 5 \Rightarrow \boxed{M = 5}$$

(c). $M = 4, \quad X_d(\Omega) = \frac{1}{M} X\left(\frac{\Omega}{M}\right) = \frac{1}{4} X\left(\frac{\Omega}{4}\right)$

$\Omega_0^d = 4\Omega_0 = 4 \times \frac{\pi}{5} = \frac{4}{5}\pi$



(d): $L = 3$

